AMTCS System – User Guide

1. System Overview

The AMTCS application is designed for automated control of temperature-resolved spectroscopic measurements, integrating multiple laboratory devices into a single workflow.

An experiment consists of a sequence of measurements performed at different temperature setpoints. A measurement is an individual measurement performed at a specific temperature.

Supported Equipment

- Synapse CCD camera (HORIBA Jobin Yvon)
- iHR-320 monochromator (HORIBA Jobin Yvon)
- CryoCon-32 temperature controller
- Raspberry Pi 5 (PLC controller)
- Windows-based PC running the User Interface (UI) application; Windows 7 is supported

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2. System Setup

Before starting the application, ensure that all devices are:

- Properly interconnected
- Powered on
- Reachable from the Windows PC (for example, using ping 192.168.50.1)
- Internet Connection Sharing must be configured on the Windows PC for remote-control functionality

Devices to connect:

- Spectrometer (iHR-320 monochromator + Synapse CCD camera)
- Temperature controller
- Raspberry Pi (PLC)
- PC with a supported version of Microsoft Windows

PART I — USER INTERFACE (UI) APPLICATION

3. Starting the UI Application

1. Launch the **User Interface (UI) application** on the PC.
2. Check the UI application console log (Fig. 1) to confirm basic connectivity with the Raspberry Pi. The PLC may already be running, or the UI application may attempt to start it automatically. If the PLC cannot be started, check the connections and retry.

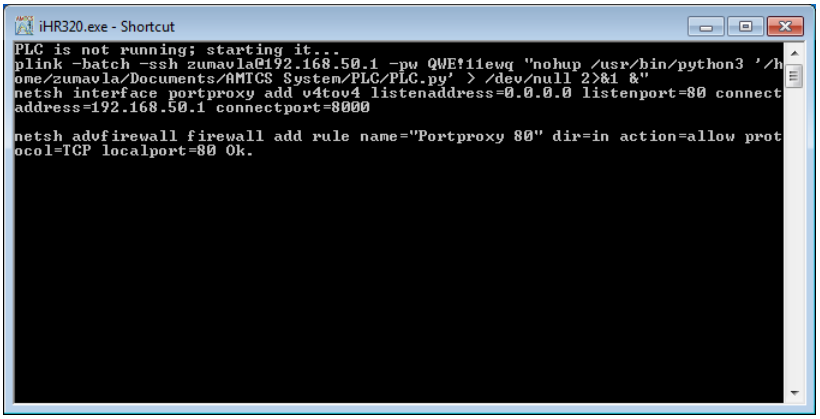


Fig. 1 Log examples in the console of the UI application at its start.

3. On the **Connectivity** screen (Fig. 2), click ***Connect to equipment***.

The system will:

- Establish communication with the connected equipment
- Initialise the required hardware components

After a successful connection, the user will be automatically redirected to the **Experiment Settings** screen.

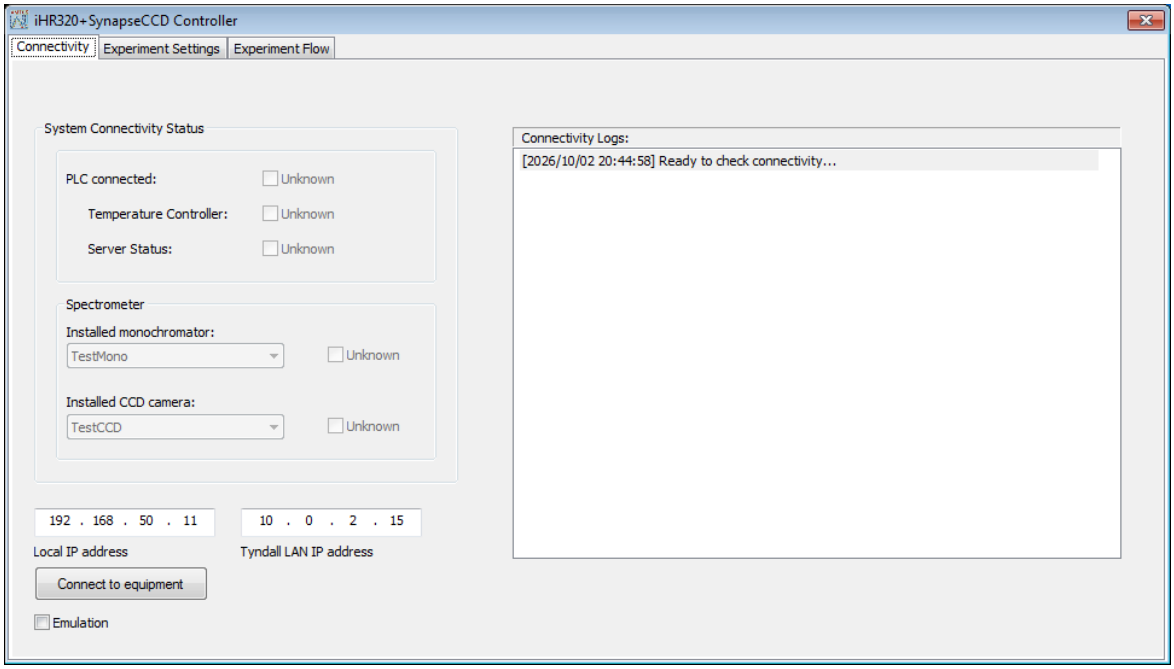


Fig. 2 UI application **Connectivity** screen with empty log list (before connecting to equipment).

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4. Experiment Setup

On the Experiment Settings screen (Fig. 3), the required experiment parameters can be specified. Default values are already preset.

Required Inputs:

- ***Save to (path)*** – select a directory where measurement results are to be saved
- ***Sample Code*** – mandatory identifier for the sample. Warning: existing data with the same Sample Code may be overwritten when the same output folder (path) is used.

Additional Parameters:

- List of temperatures
- Measurement configuration settings:
  - Diffraction grating (default grating 3)
  - Spectral range: starting wavelength, number of diffraction grating positions and wavelength exclusion range to be ignored by the *Autoexposure* feature
  - Slits (specified in micrometres)[1]
  - Acquisition settings: number of acquisitions, maximum acquisition time and *Cosmic ray removal* (select the checkbox to enable it)[2]

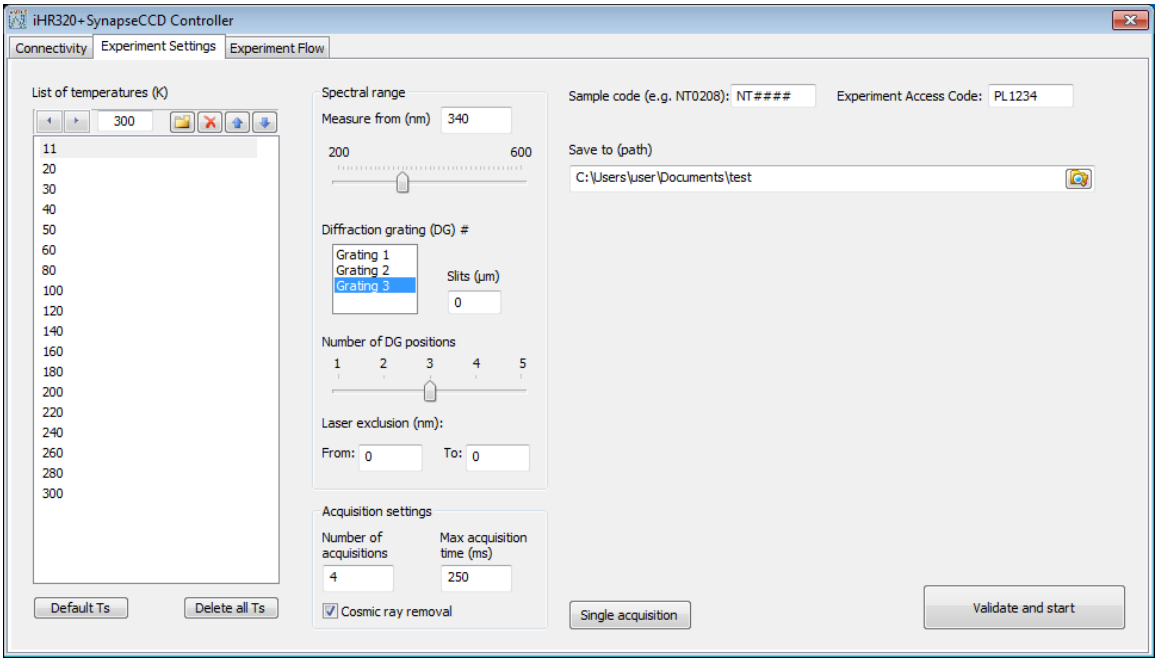


Fig. 3 **Experiment Settings** screen with default experiment parameters.

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5. Starting the Experiment

After specifying all required parameters, click ***Validate and start***

The UI application will:

- Sort the temperature list automatically, starting with the temperature point closest to the current temperature of the temperature controller
- Remove duplicate temperature points, if any
- Prepare the experiment information and send it to the PLC

The user will then be redirected to the **Experiment Flow** screen (Fig. 4).

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6. Experiment Execution

During the experiment, the system automatically:

- Moves through the temperature setpoints
- Performs a measurement at each temperature point

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[1] Note: Due to a hardware limitation, slit values below 10 µm are treated as 10 µm.
[2] Note: Autoexposure is available only when at least three acquisitions are selected and Cosmic ray removal is enabled.

- Processes, saves, and logs the measurement data

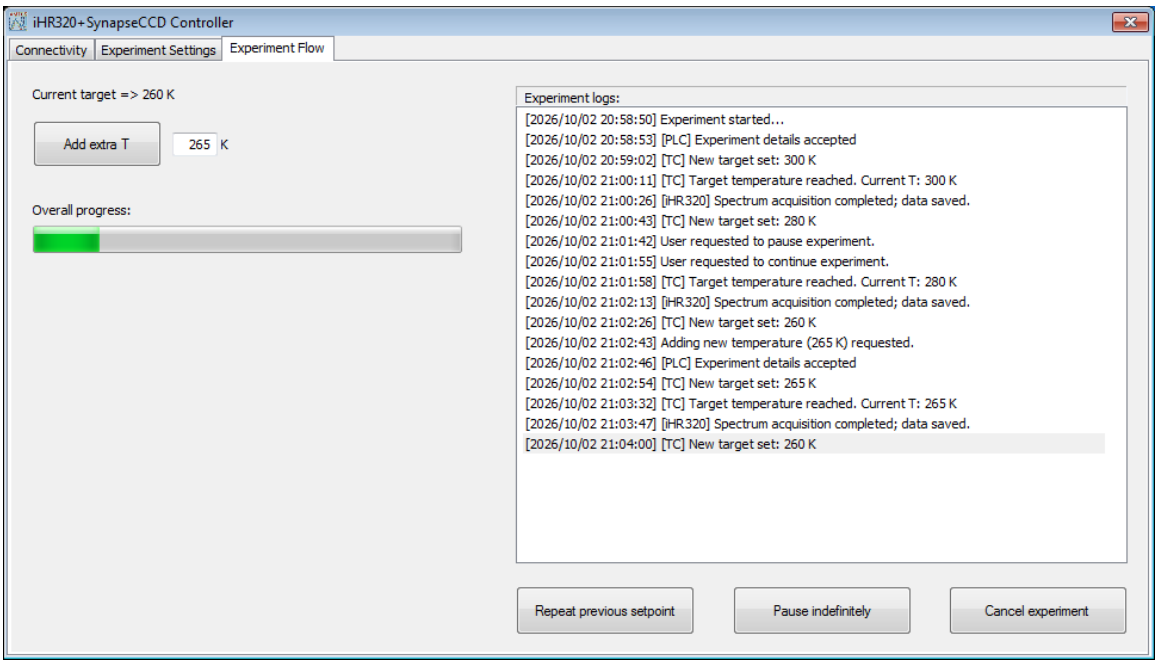


Fig. 4 Experiment Flow screen with logs showing experiment progress and user interactions.

Available Controls:

- ***Add extra T*** → Adds additional temperature point[3].
- ***Repeat previous setpoint*** → Repeats measurement at the previous temperature setpoint.
- ***Pause indefinitely/Continue*** → Suspends or resumes execution of the experiment
- ***Cancel experiment*** → Terminates the experiment early after an additional confirmation by the user.

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7. End of Experiment

After the experiment is completed, the user will be prompted to start a new experiment. If the user declines, the UI application will shut down the system, including the UI application, PLC, and remote-control server.

All measurement results generated during the experiment are saved in the selected output folder and remain there after the system is shut down.

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8. Notes and Recommendations

- Ensure all devices are connected before pressing **Connect to equipment**.
- Always provide a valid **Sample Code** and output directory (path).
- Review the temperature list before starting the experiment.
- **Important:** During active measurements, the monitors are automatically switched off to minimise stray light. Avoid unnecessary use of the mouse or keyboard while a measurement is in progress.

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[3] Note: Only values from the last measurement to the final temperature in the list are allowed as a new temperature when using Add extra T.

PART II — REMOTE CONTROL APPLICATION

The AMTCS system can be monitored and controlled remotely using the Remote Control application.

The Remote Control application provides an additional way to monitor and control an experiment without using the main UI application at the laboratory PC. It does not replace the UI application for starting a new experiment or shutting down the system.

Starting the Remote Control

1. Launch the **rc_client_portable.exe** on the PC from which you want to monitor the experiment.
2. A small terminal window will open (Fig. 5). Minimise the window and leave it running. Do not close it while using the Remote Control application.

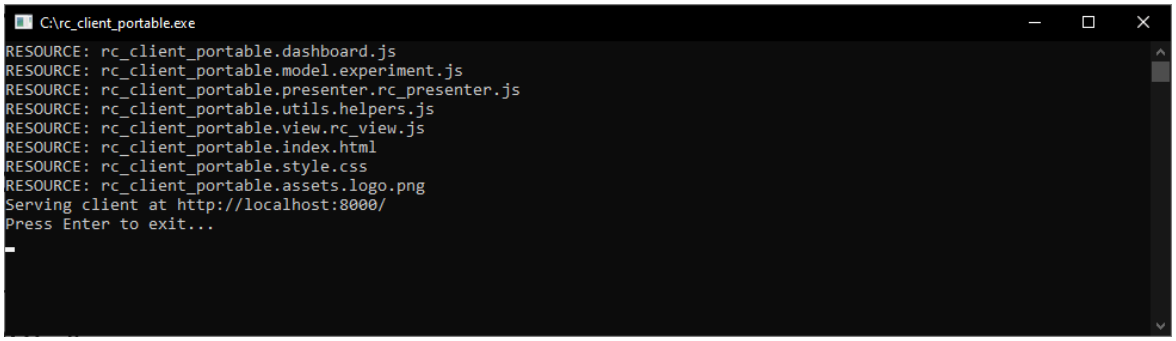


Fig. 5 Console window of the AMTCS remote control application.

3. A web browser window will open automatically with the Remote Control interface (Fig. 6).

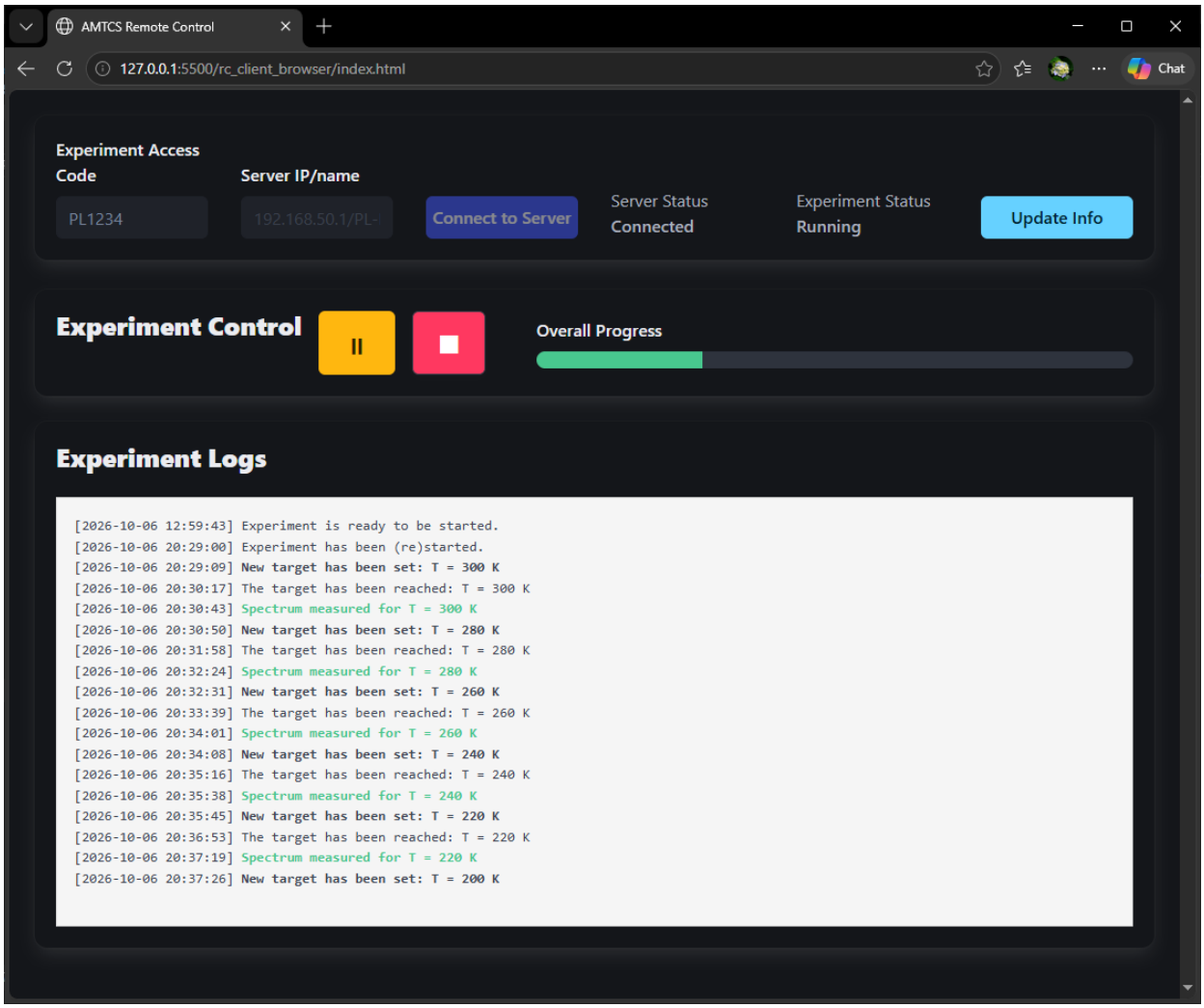


Fig. 6 Web-browser-based AMTCS remote control interface.

4. Check the Server IP/name field. If necessary, enter the server address manually. The default server name is PL-PLE[4]. If you are not sure which address to use, check the Tyndall LAN IP address displayed on the **Connectivity** screen of the UI application (see Fig. 2, for example).
5. Click ***Connect to Server***. Check **Server Status** and make sure that it changes to **Connected**.
6. Enter the **Experiment Access Code (AEC)**. The AEC must be the same as the one specified in the **Experiment Settings** screen of the UI (see Fig. 3).
7. Click ***Update Info***.
8. If the connection and AEC are correct, **Experiment Status** will change from *Unknown* to the current experiment status.
9. If the experiment is *Running*, the remote-control functions (Pause indefinitely / Continue and Cancel experiment) become available. Experiment information is then updated automatically every 5 seconds.

Status Information

The Remote Control application displays two status indicators:

- **Server Status** – shows the connection status of the Remote Control application to the AMTCS server:

[4] **PL-PLE** is the host name of the Windows PC in Tyndall's PL-Raman Spectroscopy Lab to which the HORIBA Jobin Yvon spectroscopic equipment is connected.

    - *Connected*
    - *Disconnected*
- **Experiment Status** – shows the current state of the experiment:
    - *Not started* – no experiment has been started.
    - *Unknown* – the current experiment state cannot be determined.
    - *Running* – an experiment is currently in progress.
    - *Paused* – the experiment is temporarily paused.
    - *Finished* – the experiment has completed.

Experiment Access Code (AEC)

A valid **AEC** is required to retrieve experiment information and use the full remote-control functionality.

The AEC entered in the Remote Control application must match the AEC specified in the **Experiment Settings** screen of the UI application (Fig. 3).

The default AEC is: **PL1234**

If the AEC has not been changed in the UI application, use PL1234 in the Remote Control application.

If an incorrect AEC is entered, experiment information and the full remote-control functionality may not be available.

Remote Experiment Controls

When an experiment is running, the Remote Control application provides the following controls:

- **Pause indefinitely / Continue** – pauses or resumes the experiment. This performs the same action as the corresponding control in the UI application.
- **Cancel experiment** – terminates the current experiment after an additional confirmation from the user.

Note: The Remote Control application is intended for monitoring and controlling an experiment while it is running. To shut down the AMTCS system or start a new experiment, return to the laboratory and perform the corresponding action in the UI application.